Cédric Colas

AI Research Scientist · creative bricoleur

cedriccolas.com · Google Scholar · GitHub · ccolas@mit.edu

open-ended learning | LLM agents | reinforcement learning | program synthesis | quality diversity | creative systems

I'm an AI research scientist working at the intersection of artificial intelligence and cognitive science. I study autotelic agents: systems that generate, pursue, and revise their own goals. My work combines reinforcement learning, large language models, program synthesis, quality-diversity methods, and cultural evolution to model open-ended learning in individuals and populations.

Last updated: June 8, 2026.

// Research Experience

MIT, Department of Brain and Cognitive Sciences — 2022–present
Postdoctoral Researcher in Artificial Intelligence — Cambridge, USA

- Develop agentic AI systems that generate goals, explore open-ended spaces, and learn from humans, language, and other agents.
- Combine LLMs, reinforcement learning, program synthesis, and intrinsic motivation to model and engineer adaptive behavior.
- Marie Skłodowska-Curie Postdoctoral Fellowship; hosted with the Flowers AI & CogSci Lab at Inria.
- Mentored by Joshua Tenenbaum and Jacob Andreas.

Inria, Flowers AI & CogSci Lab — 2017–2021
PhD Researcher in Artificial Intelligence — Bordeaux, France

- Developed the autotelic reinforcement learning framework: agents able to generate, represent, and pursue their own goals.
- Combined reinforcement learning, language grounding, intrinsic motivation, and quality-diversity methods to engineer agents that explore and acquire open-ended skill repertoires.
- PhD thesis: *Towards Vygotskian Autotelic Agents: Learning Skills with Goals, Language and Intrinsically Motivated Deep Reinforcement Learning.* Awarded the AFIA Best PhD Thesis in AI in France.
- Supervised by Pierre-Yves Oudeyer and Olivier Sigaud.

Uber AI Labs — Jun–Sept 2019
Research Intern — San Francisco, USA

- Worked on scaling quality-diversity algorithms to deep neuroevolution with evolution strategies (GECCO 2020).
- Supervised by Vashisht Madhavan and Jeff Clune.

// Selected Publications

CogSci 2025 & ICLR 2026	**Colas**, Mills, Prystawski, Henry, Goodman, Andreas, Tenenbaum. *Language and Experience: A Computational Model of Social Learning in Complex Tasks.* [paper]
ICML 2025	Pourcel, **Colas**, Oudeyer. *Self-Improving Language Models for Evolutionary Program Synthesis: A Case Study on ARC-AGI.* ARC Prize 2025, 2nd Paper Award. [paper]
ICML 2025	Gaven, Carta, Romac, **Colas**, Lamprier, Sigaud, Oudeyer. *MAGELLAN: Metacognitive Predictions of Learning Progress Guide Autotelic LLM Agents in Large Goal Spaces.* [paper]
NeurIPS 2023	Pourcel, **Colas**, Molinaro, Oudeyer, Teodorescu. *ACES: Generating a Diversity of Challenging Programming Puzzles with Autotelic Generative Models.* [paper]
ICML 2023	Du, Watkins, Wang, **Colas**, Darrell, Abbeel, Gupta, Andreas. *Guiding Pretraining in Reinforcement Learning with Large Language Models.* [paper]
Nat. MI 2022	**Colas**, Karch, Moulin-Frier, Oudeyer. *Language and Culture Internalization for Human-Like Autotelic AI.* [paper]
JAIR 2021	**Colas**, Karch, Sigaud, Oudeyer. *Autotelic Agents with Intrinsically Motivated Goal-Conditioned Reinforcement Learning: A Short Survey.* [paper]

NeurIPS 2020	**Colas**, Karch, Lair, Dussoux, Moulin-Frier, Dominey, Oudeyer. *Language as a Cognitive Tool to Imagine Goals in Curiosity-Driven Exploration.* [paper]
GECCO 2020	**Colas**, Huizinga, Madhavan, Clune. *Scaling MAP-Elites to Deep Neuroevolution.* [paper]
ICML 2018	**Colas**, Sigaud, Oudeyer. *CURIOUS: Intrinsically Motivated Modular Multi-Goal Reinforcement Learning.* [paper]

// Technical Strengths

Methods	Reinforcement learning, intrinsic motivation, goal-conditioned RL, LLM agents, program synthesis, evolutionary search, quality-diversity, autotelic learning, Bayesian modeling, cognitive modeling, cultural evolution, human–AI experiments.
Engineering	Python, PyTorch, JAX, large-scale experimentation, Git, LaTeX.
Domains	Open-ended learning, autotelic agents, creative AI systems, collective intelligence, social learning, language-guided exploration.

// Grants, Awards & Scientific Leadership

MSCA Fellowship	Marie Skłodowska-Curie Postdoctoral Fellowship, 3-year independent research fellowship, EUR 273,000	2022–2025
Thesis Award	Best PhD Thesis in Artificial Intelligence in France (AFIA).	2022
Corr Bayen Award	Honorary Mention, 2nd rank, European Corr Bayen Young Researcher Award in Computer Science	2022
IMOL Chair	Program chair for the Intrinsically Motivated Open-ended Learning conference	2025
NeurIPS Workshops	Co-organizer of IMOL 2023, IMOL 2024, and Language and Reinforcement Learning 2022	2022–2024
CogSci Symposium	Co-organizer of *What Should I Do Now? Goal-Centric Outlooks on Learning, Exploration, and Communication*	2024
TopiCS Editor	Co-guest editor of a special issue on *Goal Dynamics* for *Topics in Cognitive Science*, bringing together 14 papers from an interdisciplinary group of researchers.	2026
MIT Art Grant	Tangible Dreams: interactive physical neural network installation; exhibited in MIT's Stata building	2025
Aalto Visit	Aalto School of Science Visiting Researcher Programme	2026

// Mentoring

Co-supervised 14 PhD students and 5 Master's students on projects in autotelic agents, language-conditioned RL, LLM agents, program synthesis, social learning, collective intelligence, and computational cognitive science.

// Software & Creative Work

Research software	Systems for intrinsically motivated RL, language-guided goal exploration, quality-diversity, epidemiological optimization, and statistical comparison of RL algorithms. See GitHub.
Creative systems	Interactive installations, sound–taste machines, embodied AI prototypes, and visual exploration systems, including *Tangible Dreams*, *Piano Ivre*, and *Chit*. More projects.

// Education

PhD, AI	Inria, Flowers AI & CogSci Lab. Thesis: *Towards Vygotskian Autotelic Agents.* Supervised by Pierre-Yves Oudeyer and Olivier Sigaud.	2017–2021
MSc, CogSci	École Normale Supérieure, Paris. Ranked 2/48.	2016–2017
MSc, Biomed. Eng.	Imperial College London, Neurotechnology stream. Graduated with distinction.	2015–2016
Engineering Degree	CentraleSupélec, Electrical Engineering, Computer Science, and Telecommunications.	2013–2016